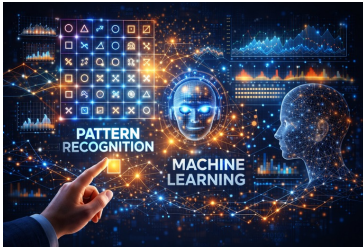


Introduction to Machine Learning
 CSL2050 - Sem 02 – 2026
 Lecture 21-22: Intro to Perceptron Learning



Avinash Sharma
 3DVisLab, CSE, IIT Jodhpur

IC: ChatGPT

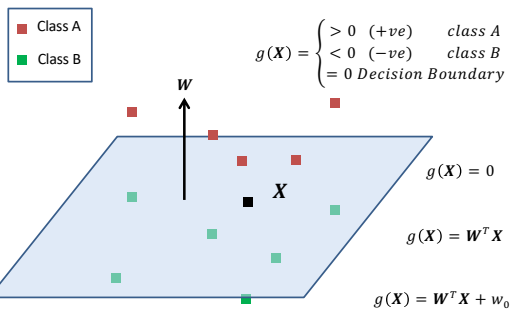
1

Linear Discriminant Functions

- Assumes a Two-class classification setup
- Decision boundary is represented explicitly in terms of components of X .
- Aim is to seek parameters of a linear discriminant function which minimize the training error.
- Why Linear ?
 - Simplest possible
 - Generalized

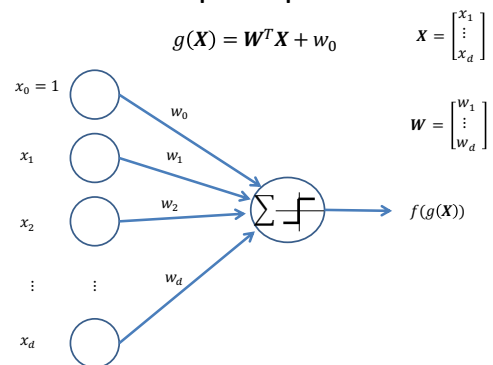
2

Linear Discriminant Functions



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The perceptron



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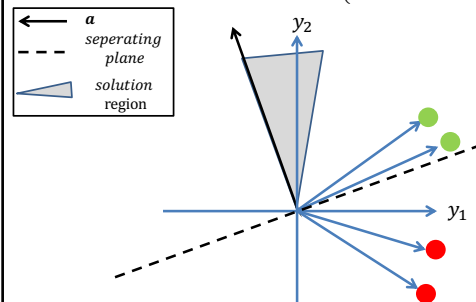
Generalized LDFs Summary

- φ can be any arbitrary mapping function that projects original data points $X \in \mathbb{R}^d$ to points $Y \in \mathbb{R}^{\hat{d}}$ where $\hat{d} \gg d$.
- The hyperplane decision surface $\hat{H}$ passes through origin.
- **Advantage:** In the mapped higher dimensional space data might be linear separable.
- **Disadvantage:** The mapping is computationally intensive and learning the classification parameters can be non-trivial (Curse of Dimensionality).

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Two-Category Linearly Separable Case

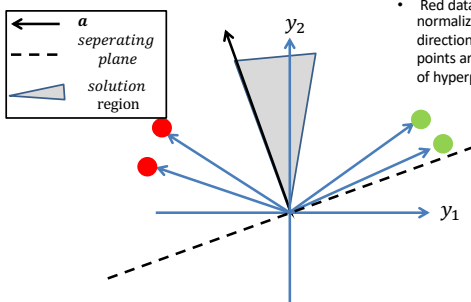
$$g(X) = \mathbf{a}^T \mathbf{Y} = \sum_{i=1}^{\hat{d}} a_i y_i = \begin{cases} > 0 & (+ve) & \text{class A} \\ < 0 & (-ve) & \text{class B} \\ = 0 & \text{Decision Boundary} \end{cases}$$



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Two-Category Linearly Separable Case

$$g(X) = \mathbf{a}^T \mathbf{Y} = \sum_{i=1}^{\hat{d}} a_i y_i > 0$$



Normalized Case

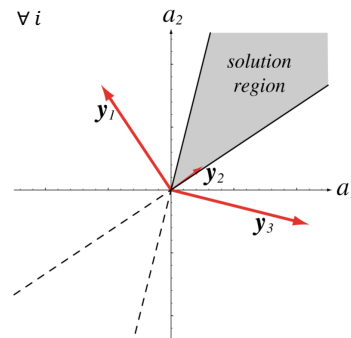
- Red data points are normalized by flipping their direction so that all data points are on the one side of hyperplane.

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Two-Category Linearly Separable Case

$$\mathbf{a}^T \mathbf{y}_i > 0 \quad \forall i$$

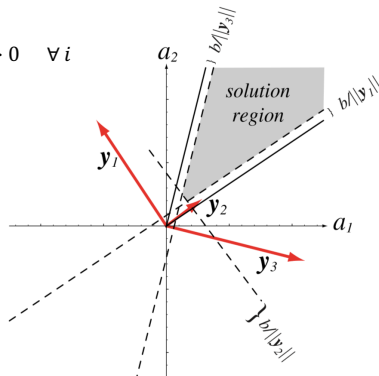
Data vector



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Two-Category Linearly Separable Case

$$\mathbf{a}^T \mathbf{y}_i \geq b > 0 \quad \forall i$$



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Learning LDF: Basic Gradient Descent

- Define a scalar function $J(\mathbf{a})$ which captures classification error for specific boundary plane described by parameter $\mathbf{a}$
- Minimize $J(\mathbf{a})$ using **gradient descent**.
 - Start with arbitrary value of $\mathbf{a}(1)$ for $k = 1$.
 - Iteratively refine estimate of $\mathbf{a}$:

$$\mathbf{a}(k+1) = \mathbf{a}(k) - \eta(k) \nabla J(\mathbf{a}(k))$$
- η is the positive scale factor known as **learning rate**
 - A too small η makes the convergence very slow
 - A too large η can diverge due to overshooting correct.

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Basic Gradient Descent Algorithm

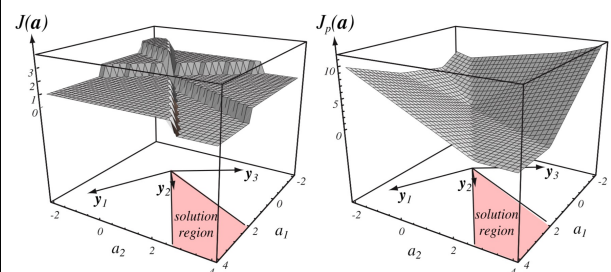
1. Initialize $\mathbf{a}, \theta$ (threshold), $\eta(\cdot), k = 0$
2. do $k = k + 1$

$$\mathbf{a}(k+1) = \mathbf{a}(k) - \eta(k) \nabla J(\mathbf{a}(k))$$
3. until $|\eta(k) \nabla J(\mathbf{a}(k))| < \theta$
4. return $\mathbf{a}$

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Perceptron Criterion Function

- Discrete v/s continuous function



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Perceptron Criterion Function

- Discrete v/s continuous function
- $J_p(\mathbf{a}) = \sum_{\mathbf{y} \in \mathcal{Y}} (-\mathbf{a}^T \mathbf{y})$ where $\mathcal{Y}$ is the set of misclassified samples
- $J_p(\mathbf{a})$ is proportional to the sum of the distances from all the miss classified samples to the decision boundary.
- Derivative of $J_p(\mathbf{a})$

$$\nabla J_p(\mathbf{a}) = \sum_{\mathbf{y} \in \mathcal{Y}} (-\mathbf{y})$$
- $\mathbf{a}(k+1) = \mathbf{a}(k) + \eta(k) \sum_{\mathbf{y} \in \mathcal{Y}_k} \mathbf{y}$

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Batch Perceptron

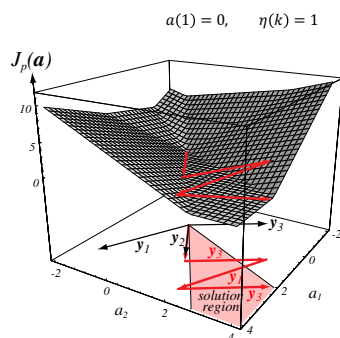
1. Initialize $\mathbf{a}, \theta$ (threshold), $\eta(\cdot), k = 0$
2. do $k = k + 1$

$$\mathbf{a}(k+1) = \mathbf{a}(k) + \eta(k) \sum_{\mathbf{y} \in \mathcal{Y}_k} \mathbf{y}$$

3. untill $|\eta(k) \sum_{\mathbf{y} \in \mathcal{Y}_k} (-\mathbf{y})| < \theta$
4. return $\mathbf{a}$

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Visualizing the Learning



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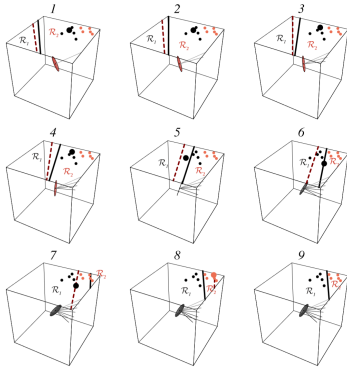
Single Sample Perceptron

1. Initialize $\mathbf{a}, k = 0$
2. do $k = \text{mod}(k + 1, n)$
 - If $\mathbf{y}^k$ is misclassified by $\mathbf{a}$ then $\mathbf{a} = \mathbf{a} + \mathbf{y}^k$
3. untill all patterns are correctly classified
4. return $\mathbf{a}$

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Single Sample Perceptron

- Geometrically, when $\mathbf{a}$ misclassifies $\mathbf{y}^k$, $\mathbf{a}$ is not on the positive side of $\mathbf{y}^k$ hyperplane $\mathbf{a}^T \mathbf{y}^k = 0$.
- Thus, adding $\mathbf{y}^k$ to $\mathbf{a}$ moves the weight vector directly towards and perhaps across this hyperplane.
- The correction is hence moving the weight vector in good direction.



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Relaxation Procedures

- These are broader class of criterion functions and associated minimization methods.

$$J_q(\mathbf{a}) = \sum_{\mathbf{y} \in \mathcal{Y}} (\mathbf{a}^T \mathbf{y})^2$$

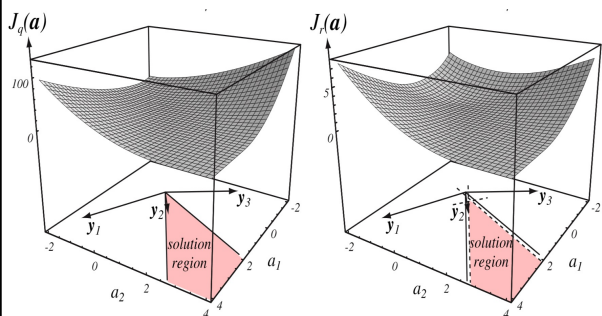
- Problems:
 - The smooth nature of J_q can lead to convergence to boundary.
 - J_q is dominated by the longest sample vector

- To address these problems, J_r is proposed as follows:

$$J_r(\mathbf{a}) = \frac{1}{2} \sum_{\mathbf{y} \in \mathcal{Y}} \frac{(\mathbf{a}^T \mathbf{y} - b)^2}{\|\mathbf{y}\|^2}$$

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Relaxation Procedures



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Single Sample Relaxation with Margin

Single Sample Relaxation with Margin

$$\nabla J_r(\mathbf{a}) = \sum_{\mathbf{y} \in \mathcal{Y}} \frac{\mathbf{a}^T \mathbf{y} - b}{\|\mathbf{y}\|^2} \mathbf{y}$$

1. Initialize $\mathbf{a}, \eta(\cdot), k = 0$

2. do $k = \text{mod}(k + 1, n)$

$$\text{if } \mathbf{a}^T \mathbf{y}^k \leq b \text{ then } \mathbf{a} = \mathbf{a} + \eta(k) \left(\frac{b - \mathbf{a}^T \mathbf{y}^k}{\|\mathbf{y}^k\|^2} \right) \mathbf{y}^k$$

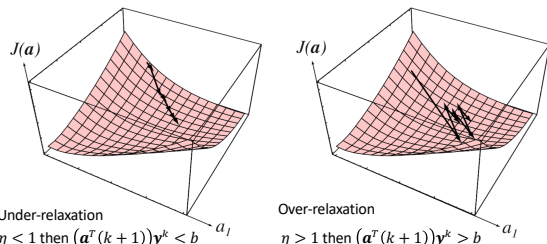
3. until $\mathbf{a}^T \mathbf{y}^k > b$ for all $\mathbf{y}^k$

4. return $\mathbf{a}$

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Over/Under-Relaxation

- $r(k) = \frac{(b - \mathbf{a}^T \mathbf{y}^k)}{\|\mathbf{y}^k\|^2} \mathbf{y}^k$ is the distance from $\mathbf{a}(k)$ to the hyperplane.



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Non Separable Behavior

- The perceptrons and relaxation procedures are "Error correcting procedures".
- Their Generalization to unseen test data not guaranteed.
- Fails to handle non-separable case.
- Many heuristic exists to handle non-separable cases:
 - Forced termination of loop after large # of iterations.
 - Annealing of η with increasing k .

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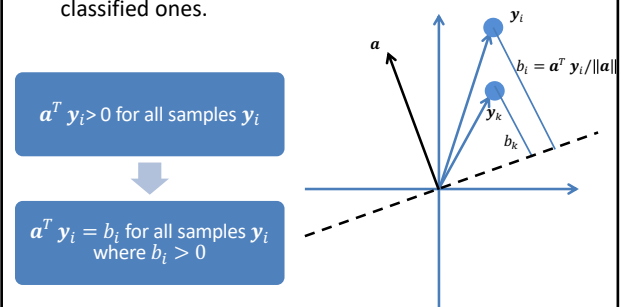
Minimum Squared Error Procedures

- MSE consider all samples instead of just miss-classified ones.
- Moved from problem of finding solution to a set of linear inequalities to a set of linear equations, i.e., $\mathbf{a}^T \mathbf{y}_i = b_i$ instead of $\mathbf{a}^T \mathbf{y}_i > 0$

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Minimum Squared Error Procedures

- MSE consider all samples instead of just miss-classified ones.



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Minimum Squared Error Procedures

- Thus $b_1, b_2, \dots, b_n$ give relative expected distances or "margins" of samples from the hyperplane.
- Should make b_i small if sample i -th point is expected to be near separating hyperplane, and make b_i larger otherwise.
- In the absence of any additional information, there are good reasons to set $b_1 = b_2 = \dots = b_n = 1$.

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Minimum Squared Error Procedures

- $Y = [\mathbf{y}_1 \ \dots \ \mathbf{y}_n]^T$ be the set of all data points where $\mathbf{y}_i = [y_{i0} \ \dots \ y_{id}]^T \in \mathbb{R}^{\hat{d}=(d+1)}$
- Let $\mathbf{a} = [a_0 \ \dots \ a_d]^T$ and $\mathbf{b} = [b_1 \ \dots \ b_n]^T$
- $Y_{n \times (d+1)} \mathbf{a}_{(d+1) \times 1} = \mathbf{b}$ (over-determined problem as $n \gg \hat{d}$)
- $\mathbf{a} = Y^{-1} \mathbf{b}$ not possible (Y is rectangular and possibly singular)
- No exact solution ! We look for approximate solution.

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Minimum Squared Error Procedures

- $Y\mathbf{a} - \mathbf{b} = \mathbf{0}$
- $\mathbf{e} = Y\mathbf{a} - \mathbf{b}$ (Error definition)
- $J(\mathbf{a}) = \|\mathbf{e}\|^2 = \|Y\mathbf{a} - \mathbf{b}\|^2$
- $\nabla J(\mathbf{a}) = 2Y^T(Y\mathbf{a} - \mathbf{b}) = \mathbf{0}$
- $Y^T Y \mathbf{a} = Y^T \mathbf{b}$
- $\mathbf{a} = (Y^T Y)^{-1} Y^T \mathbf{b} = Y^\dagger \mathbf{b}$

$Y^\dagger$ is called the pseudo-inverse matrix

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Minimum Squared Error Procedures

- $(Y^T Y)$ is a square matrix $(d+1) \times (d+1)$ and often non-singular and hence invertible.
- There could be many solutions for weight vector $\mathbf{a}$ based on choice of vector $\mathbf{b}$.
- A separating hyperplane is guaranteed if $(Y\mathbf{a} > \mathbf{0})$, i.e., $\forall i (a^T \mathbf{y}_i) > 0$
- However, we have $Y\mathbf{a} \approx \mathbf{b}$, i.e., $Y\mathbf{a} = \mathbf{b} + \boldsymbol{\varepsilon}$.
- In practice, some entries of $\mathbf{b}$ can be negative if $|b_i| < |\varepsilon_i|$ and $\varepsilon_i < 0$.

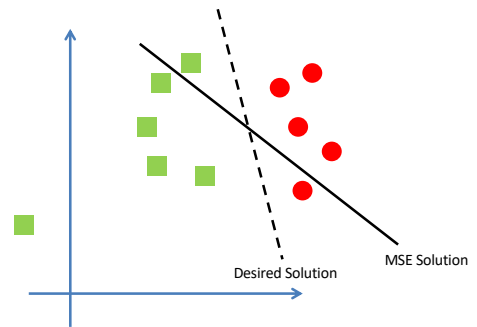
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Minimum Squared Error Procedures

- Thus, even in linearly separable case, least square solution ***a*** might not yield a separating hyperplane but a **reasonable** one.
- An arbitrary scaling of ***b*** to overcome the $-\epsilon$ values is **not helpful** as it translates to scaling up the ***a*** vector.
- However, relative difference in elements of ***b*** can help in improving the classification, especially to handle the case of outlier data points.

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Minimum Squared Error Procedures



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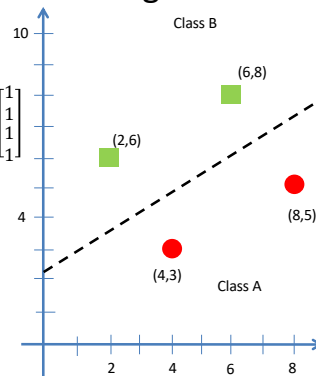
Example Walkthrough

- Class A: (8,5), (4,3)
- Class B: (2,6), (6,8)

$$Y^T = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 8 & 4 & -2 & -6 \\ 5 & 3 & -6 & -8 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\mathbf{a} = Y^\dagger \mathbf{b} = \begin{bmatrix} 1.5 \\ 0.25 \\ -0.5 \end{bmatrix}$$

$$\mathbf{y}\mathbf{a} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$



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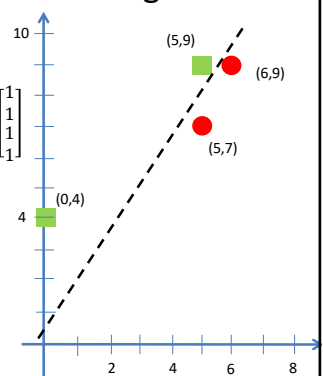
Example Walkthrough

- Class A: (6,9), (5,7)
- Class B: (5,9), (0,4)

$$Y^T = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 6 & 5 & -5 & 0 \\ 9 & 7 & -9 & -4 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\mathbf{a} = Y^\dagger \mathbf{b} = \begin{bmatrix} 2.66 \\ 1.04 \\ -0.94 \end{bmatrix}$$

$$\mathbf{y}\mathbf{a} = \begin{bmatrix} 0.43 \\ 1.28 \\ 0.60 \\ 1.11 \end{bmatrix}$$



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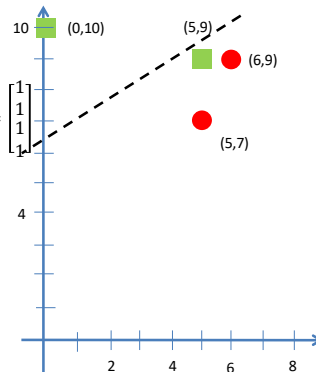
Example Walkthrough

- Class A: (6,9), (5,7)
- Class B: (5,9), (0,10)

$$Y^T = \begin{bmatrix} 1 & 1 & -1 & -1 \\ 6 & 5 & -5 & 0 \\ 9 & 7 & -9 & -10 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$a = Y^\dagger b = \begin{bmatrix} 3.21 \\ 0.15 \\ -0.43 \end{bmatrix}$$

$$Ya = \begin{bmatrix} 0.19 \\ 0.91 \\ -0.04 \\ 1.16 \end{bmatrix}$$



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Iterative Gradient Descent for MSE

- $(Y^T Y)$ is a square matrix and **often(?)** non-singular and hence invertible.
 - Singular if data points are highly correlated (rows Y are almost linear combination of each other).
- Computing inverse of $(Y^T Y)$ can be too expensive for high dimensional data (i.e., large matrix inversion).
- These issues can be avoided by adapting an iterative gradient descent solution.

$$a(k+1) = a(k) - \eta(k) \nabla J(a(k)) \text{ or}$$

$$a(k+1) = a(k) - \eta(k) Y^T (Y a(k) - b)$$

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The Widrow-Hoff /LMS Procedure

- Reduce storage requirements significantly by considering single sample sequentially, i.e., $\vec{d}$ -dimensional data point (y_i) separately.
- Prof. Bernard Widrow and Dr. Ted Hoff proved the convergence of individual "Least Mean Square (LMS) Error" minimization in the stochastic sense.

1. Initialize a, b, θ (threshold), $\eta(\cdot)$, $k = 0, i = 0$
2. do $i = \text{mod}(i + 1, n)$
3. $a(k+1) = a(k) - \eta(i) y_i (y_i^T a(k) - b_i)$
4. until $|\eta(k) y_i (y_i^T a(k) - b_i)| < \theta$
5. return a

*Use Annealing
for learning rate
 $\eta(k) = \eta(1)/k$

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Single Sample Relaxation with Margin

- Single Sample relaxation with margin

1. Initialize $a, \eta(\cdot)$, $k = 0$

2. do $k = \text{mod}(k + 1, n)$

$$\text{if } a^T y^k \leq b \text{ then } a = a + \eta(k) \frac{(b - a^T y^k)^2}{\|y^k\|^2} y^k$$

3. until $a^T y^k > b$ for all y^k

4. return a

Relaxation Procedures

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The Ho-Kashyap Procedure

- In the MSE procedure, for linearly separable case one can find a separable hyperplane ($\mathbf{a}$) iff $\mathbf{b} > \mathbf{0}$, which however is chosen arbitrarily.
- Here, the idea is to find both $\mathbf{a}$ and $\mathbf{b}$, simultaneously using the modified gradient descent procedure that minimizes $J(\mathbf{a}, \mathbf{b})$.
 - $J(\mathbf{a}, \mathbf{b}) = \|\mathbf{e}\|^2 = \|\mathbf{Y}\mathbf{a} - \mathbf{b}\|^2$
 - Fix $\mathbf{b}$ and minimize $J(\mathbf{a}, \mathbf{b})$ w.r.t. to $\mathbf{a}$
 - Fix $\mathbf{a}$ and minimize $J(\mathbf{a}, \mathbf{b})$ w.r.t. to $\mathbf{b}$
- The partial derivatives of $J(\mathbf{a}, \mathbf{b})$ will be
 - $\nabla_{\mathbf{a}} J(\mathbf{a}, \mathbf{b}) = 2\mathbf{Y}^T(\mathbf{Y}\mathbf{a} - \mathbf{b})$
 - $\nabla_{\mathbf{b}} J(\mathbf{a}, \mathbf{b}) = -2(\mathbf{Y}\mathbf{a} - \mathbf{b})$ or $(\mathbf{Y}\mathbf{a} - \mathbf{b}) = -\frac{1}{2}\nabla_{\mathbf{b}} J(\mathbf{a}, \mathbf{b}) = \mathbf{e}$

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The Ho-Kashyap Procedure

- The second step will look like

$$\mathbf{b}(k+1) = \mathbf{b}(k) - \eta(k)\nabla_{\mathbf{b}} J(\mathbf{a}(k), \mathbf{b}(k))$$
- However, during the iterative gradient descent we need to always ensure that $\mathbf{b} > \mathbf{0}$.
- This can be enforced by reducing the positive elements of $\nabla_{\mathbf{b}} J$ to zero.

$$\mathbf{b}(k+1) = \mathbf{b}(k) - \eta(k)\frac{1}{2}(\nabla_{\mathbf{b}} J - |\nabla_{\mathbf{b}} J|)^{**}$$

$$\mathbf{b}(k+1) = \mathbf{b}(k) + \eta(k)\left(-\frac{1}{2}\nabla_{\mathbf{b}} J + \frac{1}{2}|\nabla_{\mathbf{b}} J|\right)$$

$$\mathbf{b}(k+1) = \mathbf{b}(k) + \eta(k)(\mathbf{e}_k + |\mathbf{e}_k|)$$

** $|\mathbf{v}|$ is a vector with all components having absolute values of vector $\mathbf{v}$

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The Ho-Kashyap Procedure

1. Initialize $\mathbf{a}, \mathbf{b}, \eta(\cdot) < 1, k = 0, threshold(b_{min}, k_{max})$
2. do $k = k + 1$
3. $\mathbf{e}_k = (\mathbf{Y}\mathbf{a}(k) - \mathbf{b}(k))$
4. $\mathbf{b}(k+1) = \mathbf{b}(k) + \eta(k)(\mathbf{e}_k + |\mathbf{e}_k|)$
5. $\mathbf{a}(k+1) = \mathbf{Y}^+ \mathbf{b}(k+1)$
6. if $|\mathbf{e}_k| \leq b_{min}$ then return $\mathbf{a}, \mathbf{b}$ and exit
7. until $k < k_{max}$
8. Print "NO SOLUTION"

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The Ho-Kashyap Procedure

- Since $\mathbf{a}$ is determined by $\mathbf{b}$ only (step 5), the procedure can be interpreted as the one which sequentially produce margin vectors.
- Not doing steepest descent anymore, but we are still doing descent and ensure that $\mathbf{b}$ is positive.
- In case when none of the element of $\mathbf{e}_k$ vector is positive then
 - Either they all are (close to) zero which means we have solution
 - Or there is no solution
- In Linearly Separable case: solution will be found in finite step when we reach to $\mathbf{e}_k = \mathbf{0}$.
- In the Non Separable case: $\mathbf{e}_k$ will have only negative components.
 - No bound on number of iterations needed to prove the non-separability.

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Thank You